Depth-Averaged Eulerian Model for Coffee Ring Formation

Mathematical Theory and Numerical Implementation

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Abstract

This document presents a rigorous mathematical derivation of the depth-averaged Eulerian model for simulating the coffee ring effect in evaporating sessile droplets. We derive the governing equations from first principles, including mass conservation, evaporative flux, velocity fields, and particle transport. The numerical implementation is described in detail, with particular attention to stability and accuracy.

Contents

1 Introduction

When a droplet of coffee (or any colloidal suspension) dries on a solid substrate, particles accumulate at the edge, forming a characteristic ring-shaped deposit. This phenomenon, known as the *coffee ring effect*, was first systematically studied by Deegan et al. (1997).

1.1 Physical Mechanism

The coffee ring forms due to the following sequence of events:

1. The contact line (edge of the droplet) becomes **pinned** to the substrate due to surface roughness or chemical heterogeneity.
2. Evaporation occurs over the entire droplet surface, with **enhanced evaporation at the edge** due to the geometry of vapor diffusion.
3. To maintain the pinned contact line while losing volume, liquid must flow **outward** from the center toward the edge.
4. This outward flow carries suspended particles to the edge, where they **deposit** as the liquid film becomes thin.

1.2 Modeling Approach

We employ a **depth-averaged Eulerian** approach:

- **Eulerian**: Track concentration field $c(r, t)$ on a fixed grid, not individual particles.
- **Depth-averaged**: Integrate over the droplet height to obtain 2D equations in (r, t).
- **Axisymmetric**: Assume radial symmetry, reducing to 1D in r.

This approach is computationally efficient and captures the essential physics of coffee ring formation.

2 Droplet Geometry

2.1 Spherical Cap Approximation

For small contact angles $\theta \ll 1$, the droplet shape is well-approximated by a spherical cap. Consider a droplet with:

- Contact radius: R (constant for pinned contact line)
- Contact angle: $\theta(t)$ (decreases as droplet evaporates)
- Radius of curvature: $R_c = R/\sin\theta \approx R/\theta$ for small θ

Derivation of Height Profile

The spherical cap surface is described by:

$$z = R_c - \sqrt{R_c^2 - r^2}$$

For small θ, we have $R_c \gg R$, so we can expand:

$$\sqrt{R_c^2 - r^2} = R_c\sqrt{1 - \frac{r^2}{R_c^2}} \approx R_c\left(1 - \frac{r^2}{2R_c^2}\right) = R_c - \frac{r^2}{2R_c}$$

The apex height is $h_0 = R_c - \sqrt{R_c^2 - R^2} \approx R^2/(2R_c) = R\theta/2$.
Therefore, the height profile is:

$$h(r) = R_c - \sqrt{R_c^2 - r^2} - \left(R_c - \sqrt{R_c^2 - R^2}\right) \approx \frac{R^2 - r^2}{2R_c} = \frac{(R^2 - r^2)\theta}{2R}$$

Height Profile

$$h(r,t) = \frac{(R^2 - r^2)\theta(t)}{2R} \tag{1}$$

Properties:

- At center: $h(0) = R\theta/2 = h_0$ (apex height)
- At edge: $h(R) = 0$ (contact line)
- Parabolic profile in r^2

2.2 Droplet Volume

The total volume of the droplet is:

$$\begin{aligned} V &= \int_0^R 2\pi r\, h(r)\, \mathrm{d}r = \int_0^R 2\pi r \cdot \frac{(R^2 - r^2)\theta}{2R}\, \mathrm{d}r \\ &= \frac{\pi\theta}{R} \int_0^R (R^2 r - r^3)\, \mathrm{d}r = \frac{\pi\theta}{R} \left[\frac{R^2 r^2}{2} - \frac{r^4}{4}\right]_0^R \\ &= \frac{\pi\theta}{R} \left(\frac{R^4}{2} - \frac{R^4}{4}\right) = \frac{\pi\theta}{R} \cdot \frac{R^4}{4} \end{aligned} \tag{2}$$

Droplet Volume

$$V = \frac{\pi R^3 \theta}{4} \tag{3}$$

3 Contact Angle Evolution

3.1 Constant Contact Radius (CCR) Mode

For a pinned droplet, the contact radius R remains constant while the contact angle θ decreases as liquid evaporates. This is called the **CCR mode**.

3.2 Linear Approximation

Following Popov (2005), we approximate the contact angle evolution as linear in time:

Contact Angle Evolution

$$\theta(t) = \theta_0 \left(1 - \frac{t}{t_f}\right) \tag{4}$$

where:

- θ_0 = initial contact angle

- t_f = total drying time

3.3 Height Time Derivative

Time Derivative of Height

From equation (1):

$$h(r,t) = \frac{(R^2 - r^2)\theta(t)}{2R}$$

Taking the time derivative at fixed r:

$$\frac{\partial h}{\partial t} = \frac{(R^2 - r^2)}{2R}\frac{\mathrm{d}\theta}{\mathrm{d}t}$$

From equation (4):

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = -\frac{\theta_0}{t_f} = -\frac{\theta}{t_f - t}$$

Substituting:

$$\frac{\partial h}{\partial t} = \frac{(R^2 - r^2)}{2R}\left(-\frac{\theta}{t_f - t}\right) = -\frac{h}{t_f - t}$$

Height Evolution

$$\frac{\partial h}{\partial t} = -\frac{h}{t_f - t} \tag{5}$$

Note: This is negative (height decreasing) and accelerates as $t \to t_f$.

4 Evaporative Flux

4.1 Diffusion-Limited Evaporation

Evaporation is controlled by the diffusion of vapor away from the droplet surface. The vapor concentration c_v satisfies Laplace's equation in the gas phase:

$$\nabla^2 c_v = 0$$

with boundary conditions:

- At the droplet surface: $c_v = c_{sat}$ (saturated vapor concentration)
- Far from the droplet: $c_v = c_\infty = RH \cdot c_{sat}$ (ambient, with relative humidity RH)

4.2 Evaporative Flux Distribution

Hu & Larson (2002) solved this problem for a spherical cap and found that the local evaporative flux has a singularity at the contact line:

Evaporative Flux

$$J(r) = J_0\left(1 - \tilde{r}^2\right)^{-\lambda} \tag{6}$$

where:

- $\tilde{r} = r/R$ (dimensionless radial position)
- $\lambda = \dfrac{\pi - 2\theta}{2\pi - 2\theta}$ (singularity exponent)
- $J_0 = \dfrac{D_v \Delta c}{R}(0.27\theta^2 + 1.30)$ (prefactor)
- D_v = vapor diffusivity in air
- $\Delta c = c_{sat}(1 - RH)$ (vapor concentration difference)

4.3 Physical Interpretation

- For small θ: $\lambda \approx 1/2$, giving $J \sim (R-r)^{-1/2}$ near the edge.
- The flux **diverges at the contact line** because vapor can diffuse away more easily from the sharp edge.
- This edge-enhanced evaporation is the fundamental driver of the coffee ring effect.

4.4 Drying Time

The total evaporation rate is:

$$\frac{\mathrm{d}V}{\mathrm{d}t} = -\frac{1}{\rho}\int_0^R 2\pi r\, J(r)\, \mathrm{d}r$$

The drying time is found by requiring that the initial volume V_0 evaporates completely:

Drying Time

$$t_f = \frac{V_0}{\displaystyle\int_0^R 2\pi r\, J(r)/\rho\, \mathrm{d}r} = \frac{\rho V_0}{\dot{m}} \tag{7}$$

where $\dot{m}$ is the total mass evaporation rate.

5 Depth-Averaged Velocity

This is the central derivation of the model.

5.1 Mass Conservation

Consider an annular control volume between r and $r + \mathrm{d}r$. The liquid mass balance states:

Rate of height change = Net radial flux in - Evaporative loss

In differential form, the **depth-integrated continuity equation** is:

$$\frac{\partial h}{\partial t} + \frac{1}{r}\frac{\partial}{\partial r}(r h \overline{u}_r) = -\frac{J}{\rho} \tag{8}$$

where $\overline{u}_r(r,t)$ is the depth-averaged radial velocity.

5.2 Derivation of Velocity Field

Depth-Averaged Velocity

Rearrange equation (8):

$$\frac{1}{r}\frac{\partial}{\partial r}(rh\overline{u}_r) = -\frac{J}{\rho} - \frac{\partial h}{\partial t}$$

Multiply both sides by r:

$$\frac{\partial}{\partial r}(rh\overline{u}_r) = r\left(-\frac{J}{\rho} - \frac{\partial h}{\partial t}\right)$$

Integrate from 0 to r, using $\overline{u}_r(0) = 0$ (symmetry at center):

$$rh\overline{u}_r = \int_0^r r'\left(-\frac{J(r')}{\rho} - \frac{\partial h(r')}{\partial t}\right) \mathrm{d}r'$$

Substitute $\partial h/\partial t = -h/(t_f - t)$ from equation (5):

$$rh\overline{u}_r = \int_0^r r'\left(-\frac{J(r')}{\rho} + \frac{h(r')}{t_f - t}\right) \mathrm{d}r'$$

Solve for $\overline{u}_r$:

$$\overline{u}_r = \frac{1}{rh}\int_0^r \left[\frac{h(r')}{t_f - t} - \frac{J(r')}{\rho}\right] r'\,\mathrm{d}r'$$

Depth-Averaged Radial Velocity

$$\boxed{\overline{u}_r(r,t) = \frac{1}{rh}\int_0^r \left[\frac{h(r')}{t_f - t} - \frac{J(r')}{\rho}\right] r'\,\mathrm{d}r'} \qquad (9)$$

5.3 Physical Interpretation of Velocity Terms

1. **Height release term** ($+h/(t_f - t)$): As the interface descends, liquid is "squeezed out" and must flow outward. This term is positive and drives outward flow.
2. **Evaporation sink term** ($-J/\rho$): Evaporation removes liquid, reducing the amount that needs to flow outward. This term is negative.
3. **Net result**: Since $\int h \cdot r\,\mathrm{d}r$ (total volume) exceeds $\int J \cdot r\,\mathrm{d}r$ (local evaporation) for most of the droplet interior, the velocity is **positive (outward)** everywhere.

5.4 Singular Behavior

Near the contact line ($r \to R$):

- $h \to 0$ (film thickness vanishes)
- The integral in the numerator remains finite
- Therefore $\overline{u}_r \sim 1/h \sim (R - r)^{-1} \to \infty$

Near the end of drying ($t \to t_f$):

- $t_f - t \to 0$
- The height release term $h/(t_f - t) \to \infty$
- This creates a "**rush hour**" effect where velocity diverges

6 Particle Transport Equation

6.1 Conservation of Suspended Particles

Let $c(r,t)$ be the depth-averaged particle concentration (particles per unit volume of liquid). The conservation equation for suspended particles is:

$$\frac{\partial(hc)}{\partial t} + \frac{1}{r}\frac{\partial}{\partial r}(rh\overline{u}_r c) = \frac{1}{r}\frac{\partial}{\partial r}\left(rhD\frac{\partial c}{\partial r}\right) - S_{dep}$$

6.2 Simplification Using Liquid Mass Conservation

Particle Transport Equation

Expand the left-hand side:

$$h\frac{\partial c}{\partial t} + c\frac{\partial h}{\partial t} + h\overline{u}_r\frac{\partial c}{\partial r} + c\cdot\frac{1}{r}\frac{\partial}{\partial r}(rh\overline{u}_r) = \frac{D}{r}\frac{\partial}{\partial r}\left(rh\frac{\partial c}{\partial r}\right) - S_{dep}$$

Group the terms involving c:

$$h\frac{\partial c}{\partial t} + h\overline{u}_r\frac{\partial c}{\partial r} + c\underbrace{\left[\frac{\partial h}{\partial t} + \frac{1}{r}\frac{\partial}{\partial r}(rh\overline{u}_r)\right]}_{=-J/\rho \text{ from eq. (8)}} = \frac{D}{r}\frac{\partial}{\partial r}\left(rh\frac{\partial c}{\partial r}\right) - S_{dep}$$

Divide by h:

$$\frac{\partial c}{\partial t} + \overline{u}_r\frac{\partial c}{\partial r} = \frac{D}{rh}\frac{\partial}{\partial r}\left(rh\frac{\partial c}{\partial r}\right) + \frac{cJ}{\rho h} - \frac{S_{dep}}{h}$$

Particle Concentration Equation

$$\boxed{\frac{\partial c}{\partial t} + \overline{u}_r\frac{\partial c}{\partial r} = \frac{D}{rh}\frac{\partial}{\partial r}\left(rh\frac{\partial c}{\partial r}\right) + \frac{cJ}{\rho h} - \frac{S_{dep}}{h}} \tag{10}$$

Terms:

Term	Physical Meaning
$\partial c/\partial t$	Local accumulation
$\overline{u}_r\partial c/\partial r$	Advection (transport by flow)
$\frac{D}{rh}\partial_r(rh\partial_r c)$	Diffusion (Brownian motion)
$cJ/(\rho h)$	Concentration by evaporation
S_{dep}/h	Loss by deposition

6.3 Evaporation Concentration Effect

The term $cJ/(\rho h)$ represents the **concentration increase** due to evaporation:

- Solvent (water) evaporates, but particles remain
- Concentration increases even without advection
- This effect is strongest where h is small (near the edge)

7 Deposition Model

7.1 Phenomenological Approach

Particles deposit on the substrate when the film becomes thin enough for them to reach the surface. We model this with a rate that depends exponentially on the local film height:

Deposition Rate

$$S_{dep} = k_{dep} \cdot c \cdot \exp\left(-\frac{h}{h^*}\right) \tag{11}$$

where:

- k_{dep} = deposition rate constant
- h^* = characteristic height for deposition
- The exponential factor is ≈ 1 when $h \ll h^*$ and ≈ 0 when $h \gg h^*$

7.2 Deposited Mass Evolution

The deposited particle density $\sigma(r,t)$ (particles per unit area) evolves as:

Deposition Evolution

$$\frac{\partial \sigma}{\partial t} = S_{dep} = k_{dep} \cdot c \cdot \exp\left(-\frac{h}{h^*}\right) \tag{12}$$

8 Dimensionless Analysis

8.1 Characteristic Scales

Quantity	Scale	Expression
Length	R	Contact radius
Height	$h_0 = R\theta_0/2$	Initial apex height
Time	t_f	Drying time
Velocity	$U = R/t_f$	Characteristic velocity
Concentration	c_0	Initial concentration

8.2 Péclet Number

The Péclet number compares advection to diffusion:

$$Pe = \frac{UR}{D} = \frac{R^2}{Dt_f} \tag{13}$$

For typical coffee ring conditions:

- $R \sim 1$ mm, $t_f \sim 100$ s, $D \sim 10^{-12}$ m^2/s (1 μm particles)
- $Pe \sim 10^4 - 10^6$
- **Advection strongly dominates diffusion**

9 Numerical Implementation

9.1 Spatial Discretization

The radial domain $[0, R]$ is discretized with N_r points:

$$r_i = \epsilon R + i\Delta r, \quad i = 0, 1, \ldots, N_r - 1$$

where $\epsilon \ll 1$ is a small offset to avoid the singularity at $r = 0$.

9.2 Velocity Calculation

The integral in equation (9) is computed using the trapezoidal rule:

$$I_i = \int_0^{r_i} \left[\frac{h(r')}{t_f - t} - \frac{J(r')}{\rho}\right] r' \, \mathrm{d}r' \approx \sum_{j=0}^{i-1} \frac{1}{2} (f_j + f_{j+1}) \, \Delta r$$

where $f_j = \left[\frac{h_j}{\tau} - \frac{J_j}{\rho}\right] r_j$ and $\tau = t_f - t$.

Then:

$$\overline{u}_{r,i} = \frac{I_i}{r_i h_i}$$

9.3 Advection (Upwind Scheme)

Since $\overline{u}_r > 0$ (outward flow), we use upwind differencing:

$$\overline{u}_r \frac{\partial c}{\partial r} \approx \overline{u}_{r,i} \cdot \frac{c_i - c_{i-1}}{\Delta r}$$

Upwind Scheme

The upwind scheme is stable for outward flow and prevents oscillations:

- If $\overline{u}_{r,i} \geq 0$: use backward difference $(c_i - c_{i-1})/\Delta r$
- If $\overline{u}_{r,i} < 0$: use forward difference $(c_{i+1} - c_i)/\Delta r$

9.4 Diffusion (Central Difference)

The diffusion term is discretized as:

$$\frac{D}{r_i h_i} \frac{\partial}{\partial r}\left(rh\frac{\partial c}{\partial r}\right) \approx \frac{D}{r_i h_i \Delta r}\left[r_{i+1/2} h_{i+1/2} \frac{c_{i+1} - c_i}{\Delta r} - r_{i-1/2} h_{i-1/2} \frac{c_i - c_{i-1}}{\Delta r}\right]$$

9.5 Time Integration

Explicit Euler method with adaptive time stepping:

$$c_i^{n+1} = c_i^n + \Delta t \cdot \mathrm{RHS}_i^n$$

9.6 CFL Stability Condition

The time step must satisfy:

$$\Delta t < \min\left(\frac{\Delta r}{\max|\overline{u}_r|}, \frac{\Delta r^2}{2D}, 0.05(t_f - t)\right) \tag{14}$$

9.7 Boundary Conditions

Location	Condition
$r = 0$	Symmetry: $\partial c/\partial r = 0$
$r = R$	Outflow: extrapolate or zero gradient

10 Summary of Equations

Complete System of Equations

$$h(r,t) = \frac{(R^2 - r^2)\theta(t)}{2R} \tag{15}$$

$$\theta(t) = \theta_0 \left(1 - \frac{t}{t_f}\right) \tag{16}$$

$$J(r) = J_0 \left(1 - \frac{r^2}{R^2}\right)^{-\lambda}, \quad \lambda = \frac{\pi - 2\theta}{2\pi - 2\theta} \tag{17}$$

$$\overline{u}_r(r,t) = \frac{1}{rh} \int_0^r \left[\frac{h(r')}{t_f - t} - \frac{J(r')}{\rho}\right] r' \, \mathrm{d}r' \tag{18}$$

$$\frac{\partial c}{\partial t} + \overline{u}_r \frac{\partial c}{\partial r} = \frac{D}{rh} \frac{\partial}{\partial r}\left(rh \frac{\partial c}{\partial r}\right) + \frac{cJ}{\rho h} - \frac{S_{dep}}{h} \tag{19}$$

$$\frac{\partial \sigma}{\partial t} = k_{dep} \cdot c \cdot \exp\left(-\frac{h}{h^*}\right) \tag{20}$$

11 Why the Coffee Ring Forms

The model explains coffee ring formation through the following chain of causation:

1. **Pinned contact line**: $R =$ constant throughout drying.
2. **Enhanced edge evaporation**: $J(r) \sim (R - r)^{-1/2}$ diverges at the edge.
3. **Mass conservation**: To maintain the pinned contact line while volume decreases, liquid must flow outward: $\overline{u}_r > 0$.
4. **Particle advection**: The outward flow carries particles toward the edge.
5. **Thin film at edge**: As $h \to 0$ near $r = R$, particles deposit.
6. **Result**: Particles accumulate at $r \approx R$, forming a ring.

12 Limitations

The depth-averaged model has several limitations:

1. **No vertical structure**: Cannot capture recirculation cells or height-dependent velocity profiles.

2. **No Marangoni effects**: Surface tension gradients (thermal or solutal) are not included.
3. **Phenomenological deposition**: The deposition model is empirical, not derived from first principles.
4. **No particle-particle interactions**: Assumes dilute suspension.
5. **Small contact angle assumption**: The spherical cap approximation breaks down for $\theta > 40^\circ$.

To address limitation (2), we develop the two-layer Marangoni model in a separate document.

13 References

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